Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering Nagpur
(An autonomous institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur-441 110
NAAC Accredited with 'A++' Grade

Handout

23GE1101: Calculus and Vectors
Session 2024-25 (ODD Term)

Course Teacher : **Dr. (Mrs.) Malabika Adak**
Office : **SC 3-02 Third Floor**
Office Hours : **Monday- Friday 10.00 A.M.to 5.00 P.M; Saturday 10.00 A.M.to3.00 P.M**
Phone : **8793435532**
Email : **9935@ycce.in**

1. **Vision Mission Statement**

Institute Vision Mission Statement

VISION:

To become the most preferred institution providing innovative, research and value based, professional education for the society at large.

MISSION:

YCCE is committed to

- Attract best talent and create learning ambience.
- Practice Innovative teaching-learning & research.
- Integrate Industry-Institute Collaborations.
- Nurture students towards holistic development and choicest career.

Department of Basic Sciences

VISION:

To be a well-recognized center with strong foundational focus on basic sciences and humanities to develop budding professionals.

MISSION:

To develop scientific temperament and mathematical aptitude for solving inter-disciplinary engineering problems with excellent communication skills and social values in a vibrant environment.

2. **Course Objectives**

This course will teach various methods for solving first order and higher order differential equations and its applications. Also, it will introduce concepts of Complex numbers and complex variables.

3. **Course Outcome**: After completion of the course, student will demonstrate the ability to

CO1 Apply the knowledge of differentiation to solve the Engineering problems.
CO2 Determine the derivatives of functions of several variables and develop the relations among the derivatives of variables.
CO3 Apply the knowledge of Beta and Gamma functions to find area, volume and mass.

CO 4 Discuss Calculus of Scalar and vector point function and use appropriate theorems to evaluate integrals of functions of single and multiple variables.

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4. **CO – PO Mapping**:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	2	2								1		1
CO 2	2	2								1		1
CO3	2	2	2							1		1
CO 4	2	2	2							1		1

5. Mapped Program Outcomes: 1,2,3,10,12

Program outcomes
1. Engineering knowledge: Apply the knowledge of mathematics, Science, Engineering fundamentals and an Engineering specialization to the solution of complex Engineering problems.
2. Problem Analysis: Identify, formulate, review research literature and analyze complex Engineering problems reaching substantiate conclusions using first principle of mathematics, natural sciences and Engineering sciences.
3. Design/development of solutions: Design solution for complex Engineering problems and design system components or process that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal and environmental considerations.
10. Communication: Communicate effectively on complex Engineering activities with the Engineering community and with society at large such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

6. **Grading:** Grading is based on the following components that are weighted as described below

SN	Criteria	Distribution of marks(%)
1	TA-1 Proctored Quizzes/Online tests	5
2	Think pair Share and Model based learning method	12
3	Class performance Attendance% **[91 to 100 % - 3 Marks, 81 – 90 % - 2Marks, 75 – 80 % - 1 Mark, <75 % - Detention / 0**	3
4	Mid Semester Examinations-I	30
5	Mid Semester Examinations-II	
6	End Semester Examination	50

7. **Syllabus and Mark Distribution Unit Wise**

22CT201	Engineering Mathematics 1	L=3	T=0	P=0	CREDITS = 3
EVALUATION SCHEME					
MSE – I	MSE – II	TA	ESE	TOTAL	ESE DURATION
15	15	20	50	100	3 Hr

Unit	Syllabus	MSE-I	MSE-II	ESE
I	**Differential Calculus I** Successive differentiation, n^{th} derivative of rational function, Trigonometrical transformations, n^{th} derivative of the product of two functions (Leibnitz's theorem), Taylor's theorem, Use of Maclaurin's theorem for one variable, standard expansions, Examples on Taylor's Theorem (Contemporary issues related to Topic) (6 hours)	5	- -	07
II	**Differential Calculus II** Definitions of Curvature, Radius of curvature for cartesian curves, Centre of curvature, Circle of curvature, Procedure for tracing the cartesian curve, Important points (singular points, Multiple points, Double points, Node, Cusp), Problems on tracing of curve. (Contemporary issues related to Topic) [7 hrs]	5	- -	07
III	**Partial Differentiation**	5	….	06

	Functions of several variables, First and higher order derivatives, Homogeneous functions, Euler's theorem on homogeneous function, Chain rule and total differential coefficient of composite functions. (Contemporary issues related to Topic) [6 Hrs]			
IV	**Integral Calculus** **G**amma function, Reduction formula, Beta function, Properties of Beta function(without proof), Relation between Beta and Gamma functions, Double and triple integrals and its applications. (Contemporary issues related to Topic) [7 Hrs]	**- -**	**5**	**07**
V	**Vector Calculus** Vector differentiation, Gradient, Divergence and Curl, Directional derivatives with physical interpretation, Solenoidal and irrotational motions, vector fields. (Contemporary issues related to Topic) [6 Hrs]	**- -**	**5**	**07**
VI	**Vector Integration & Applications** Vector integration: Line, surface and volume integrals, Statement of Stoke's theorem, Gauss divergence theorem and Green's theorem (without proof), Simple applications of these theorems. (Contemporary issues related to Topic) [7 Hrs]	**- -**	**5**	**06**

8. Teaching Plan

Lect. No.	Unit No.	Topic Cover	Date
1	**I**	Successive differentiation, , ,	16-09-2024
2	**I**	n^{th} derivative of rational function	18-09-2024
3	**I**	Trigonometrical transformations	19-09-2024
4	**I**	n^{th} derivative of the product of two functions (Leibnitz's theorem)	20-09-2024
5	**I**	Taylor's theorem, Use of Maclaurin's theorem for one variable, standard expansions	23-09-2024
6	**I**	Example of Maclaurin's Theorem	25-09-2024
7	**I**	Examples on Taylor's Theorem	26-09-2024
8	**I**	Examples on Taylor's Theorem	27-09-2024

9	**II**	Definitions of Curvature, Radius of curvature for cartesian curves	03-10-2024
10	**II**	Centre of curvature, Circle of curvature	04-10-2024
11	**II**	Problems on curvature	07-10-2024
12		Problems on curvature	09-10-2024
13	**II**	Procedure for tracing the cartesian curve	10-10-2024
14	**II**	Important points (singular points, Multiple points, Double points, Node, Cusp),.	11-10-2024
15	**II**	Problems	14-10-2024
16	**III**	Functions of several variables, First and higher order derivatives	16-10-2024
17	**III**	Homogeneous functions	17-10-2024
18	**III**	Problems	18-10-2024
19	**III**	Euler's theorem on homogeneous function	21-10-2024
20	**III**	Chain rule and total differential coefficient of composite functions.	23-10-2024
21	**III**	Mixed problems	25-10-2024
22	**III**	Jacobian	27-10-2024
23	**III**	Differentiation Under integral sign	04-11-2024
24	**III**	Innovative method	06-11-2024
25	**IV**	**G**amma function, , Relation between Beta and Gamma functions,	07-11-2024
26	**IV**	Reduction formula	08-11-2024
27	**IV**	Beta function	18-11-2024
28	**IV**	Properties of Beta function(without proof)	20-11-2024
29	**IV**	Double integrals and its applications.	21-11-2024
30	**IV**	Triple integrals and its applications.	22-11-2024
31	**IV**	Problems	25-11-2024
32	**V**	Vector differentiation	09-12-2024
33	**V**	Gradient	11-12-2024
34	**V**	Divergence	12-12-2024
35	**V**	Curl	13-12-2024
36	**V**	Directional derivatives with physical interpretation	14-12-2024
37	**V**	Solenoidal vector fields	16-12-2024
38	**V**	Irrotational motions, vector fields	18-12-2024
39	**V**	Problems	19-12-2024
40	**VI**	Vector integration: Line integrals	20-12-2024
41	**VI**	Activity	21-12-2024

42	**VI**	surface and volume integrals	23-12-2024
43	**VI**	Gauss divergence theorem	26-12-2024
44	**VI**	Green's theorem	27-12-2024
45	**VI**	Stoke's theorem	28-12-2024

9. Examinations:

1. **QUIZZES / ONLINE TESTS**: (05 MARKS): There will be a TWO proctored on campus online tests / Quizzes at the time of MSE examinations.
2. TEACHERS ASSESSMENT (12 MARKS) : AS GIVEN IN Teacher's Assessment Plan
3. **ASSIGNMENTS:** The assignment can be in the form of Question Paper numerical tasks, innovative teaching method. The assignments are to be handwritten and to be submitted in PDF format. The student must write down Name, Section, Roll No, Registration No. on each page on "Right Top Corner".

4. **SCHEDULED EXAMINATIONS:** There will be 3 scheduled exams during semester. There will be TWO Mid Sem Examinations (MSE) and shall be compulsory for evaluation of course assessment marks. The schedule of these examinations is given in the Academic Calendar for the session published by the Dean (Academic matters)

5. MSE #1 : On the basis of Unit 1 , 2 and 3 of the syllabus
6. MSE #2 : On the basis of unit 4, 5 and 6 of the syllabus.
7. REMSE : In lieu of **at the most one MSE** in which student remained absent on the basis of All Unit 1, 2, 3, 4, 5 and 6 of the syllabus
8. ESE : On the basis of Units 1 to 6 of the syllabus

10. Advance Topics Planned:

S.N.	Beyond Syllabus Topic
1	Envelops

11. Innovative Teaching Method Planned:

S.N.	Innovative Teaching Method
1	Group Assignment-Think Pair Share
2.	Model based learning method

YCCE E-LIBRARY Book Link and MOOC Link:

SN	Title

1	https://nptel.ac.in/courses/111/106/111106146/
2	https://nitkkr.ac.in/docs/5-Multiple%20Integrals%20and%20their%20Applications.pdf

12. Text Books

SN	Author Name	Title	Publication
1.	Bali Iyengar	Text Book of Engineering Mathematics	Laxmi Publications(P) Ltd.
2.	H.K.Dass	Engineering Mathematics	S.Chand and Company Pvt. Ltd
3.	H.K.Dass	Advanced Engineering Mathematics	S.Chand and Company Pvt. Ltd.

13. Reference Book

SN	Author Name	Title	Publication
1.	J.N.Wartikar and P.N.Wartikar	A Text Book of Applied Mathematics Vol. 1	Pune Vidyarthi Griha Prakashan, Pune
2.	J.N.Wartikar and P.N.Wartikar	A Text Book of Applied Mathematics Vol. 2	Pune Vidyarthi Griha Prakashan, Pune
3.	B.S. Grewal	Higher Engineering Mathematics	Khanna Publications

Name of Course Teacher: **Dr. Malabika Adak**	**Head, Department of Applied Mathematics and Humanities** **Dr. M. G. Dhote**